



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Final Presentation

Group #17: Patent Miner

List of team members:

Christian Casteel

CJ Vines

Kosi Ezewudo

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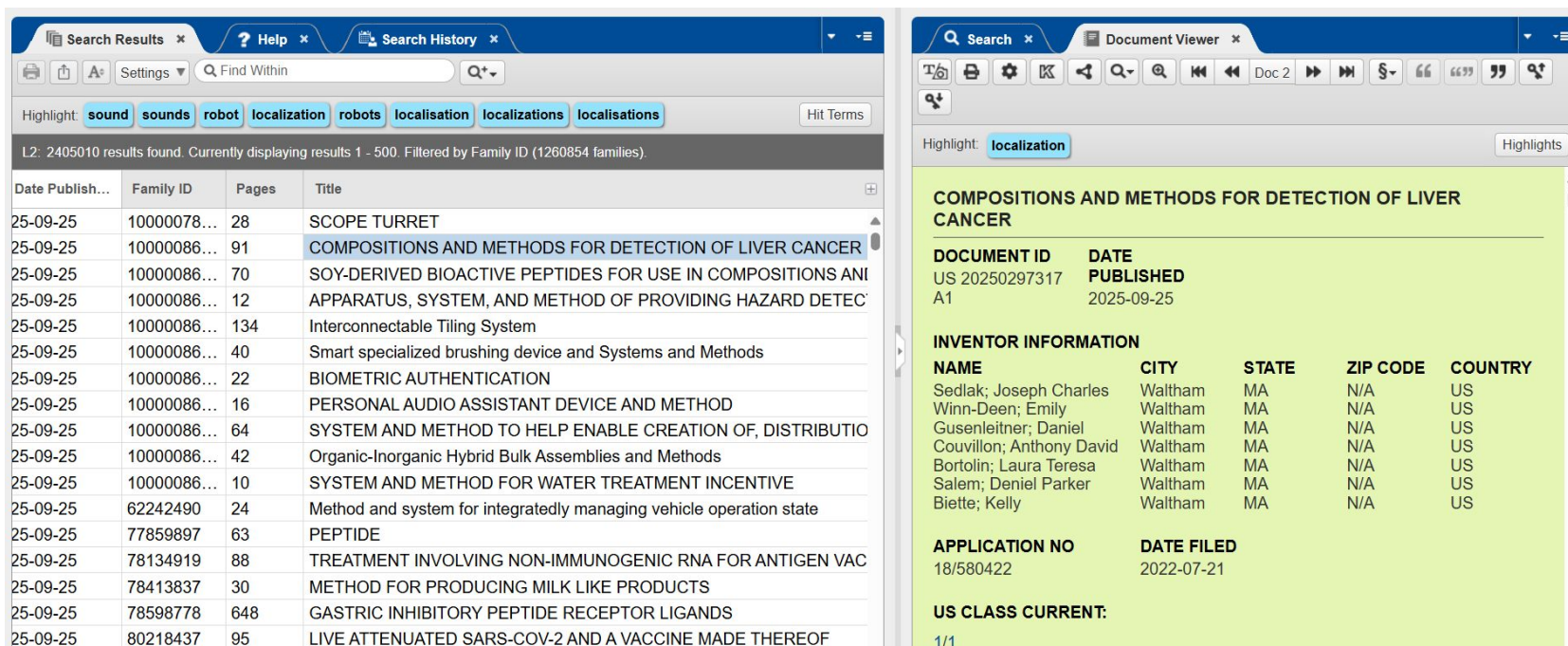
Project Summary

Problem statement:

- The current keyword search implemented by the United States Patent and Trademark Office (USPTO) fails to capture contextual meaning, leading to inaccuracy of search results and expensive billing rates for litigation professionals.

Solution:

- Development of a patent database mining tool that optimizes the patent search process by retrieving contextually relevant patents faster and more accurately compared to a traditional database.



The screenshot displays a patent search interface. The left pane shows search results for the keyword 'localization'. The right pane shows the detailed view of the selected patent, 'COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER'.

Search Results (Left Pane):

Date Publish...	Family ID	Pages	Title
25-09-25	10000078...	28	SCOPE TURRET
25-09-25	10000086...	91	COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER
25-09-25	10000086...	70	SOY-DERIVED BIOACTIVE PEPTIDES FOR USE IN COMPOSITIONS ANI
25-09-25	10000086...	12	APPARATUS, SYSTEM, AND METHOD OF PROVIDING HAZARD DETEC
25-09-25	10000086...	134	Interconnectable Tiling System
25-09-25	10000086...	40	Smart specialized brushing device and Systems and Methods
25-09-25	10000086...	22	BIOMETRIC AUTHENTICATION
25-09-25	10000086...	16	PERSONAL AUDIO ASSISTANT DEVICE AND METHOD
25-09-25	10000086...	64	SYSTEM AND METHOD TO HELP ENABLE CREATION OF, DISTRIBUTIO
25-09-25	10000086...	42	Organic-Inorganic Hybrid Bulk Assemblies and Methods
25-09-25	10000086...	10	SYSTEM AND METHOD FOR WATER TREATMENT INCENTIVE
25-09-25	62242490	24	Method and system for integratedly managing vehicle operation state
25-09-25	77859897	63	PEPTIDE
25-09-25	78134919	88	TREATMENT INVOLVING NON-IMMUNOGENIC RNA FOR ANTIGEN VAC
25-09-25	78413837	30	METHOD FOR PRODUCING MILK LIKE PRODUCTS
25-09-25	78598778	648	GASTRIC INHIBITORY PEPTIDE RECEPTOR LIGANDS
25-09-25	80218437	95	LIVE ATTENUATED SARS-COV-2 AND A VACCINE MADE THEREOF

Patent Details (Right Pane):

COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER

DOCUMENT ID US 20250297317
DATE PUBLISHED 2025-09-25

INVENTOR INFORMATION

NAME	CITY	STATE	ZIP CODE	COUNTRY
Sedlak; Joseph Charles	Waltham	MA	N/A	US
Winn-Deen; Emily	Waltham	MA	N/A	US
Gusenleitner; Daniel	Waltham	MA	N/A	US
Couvillon; Anthony David	Waltham	MA	N/A	US
Bortolin; Laura Teresa	Waltham	MA	N/A	US
Salem; Deniel Parker	Waltham	MA	N/A	US
Biette; Kelly	Waltham	MA	N/A	US

APPLICATION NO 18/580422
DATE FILED 2022-07-21

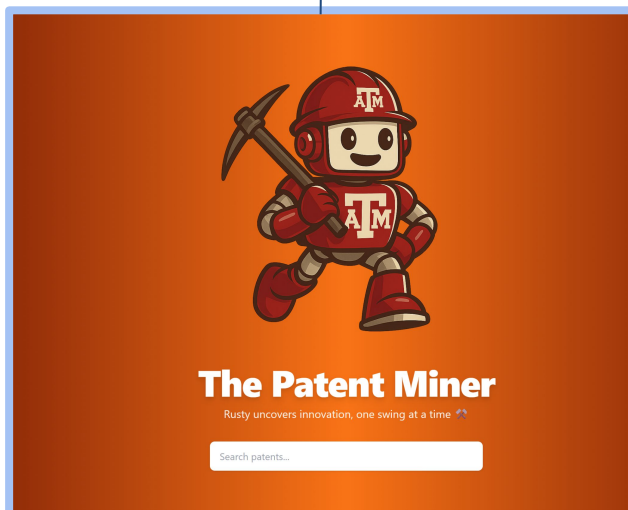
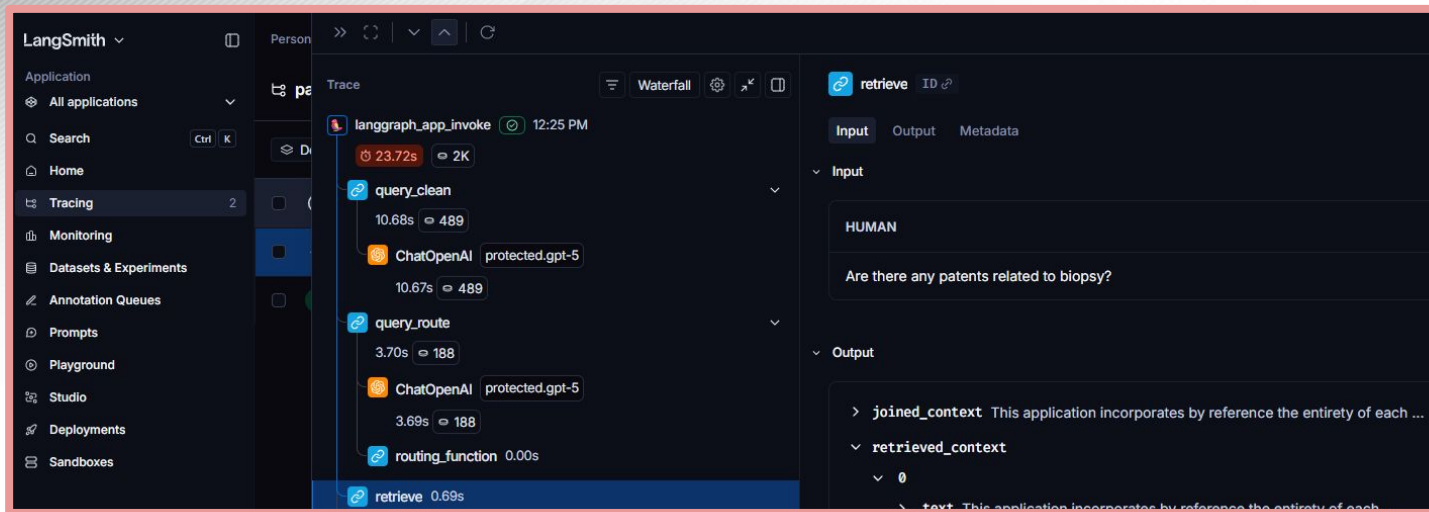
US CLASS CURRENT: 1/1



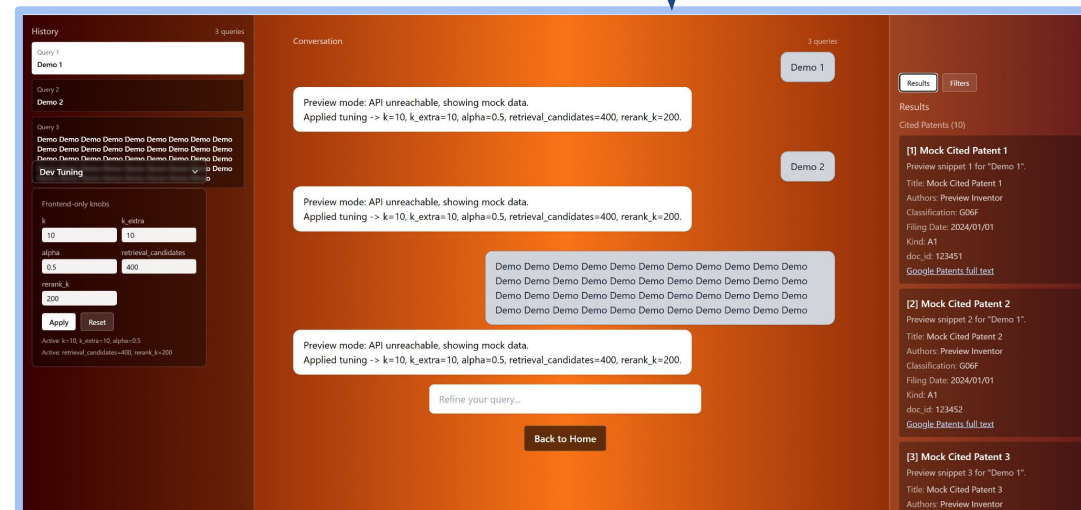
System Overview

Kosi Ezewudo

CJ Vines



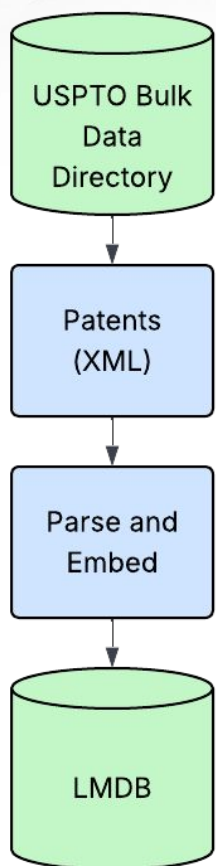
Christian Casteel



Embedding Pipeline and Storage - Embedding/Validation

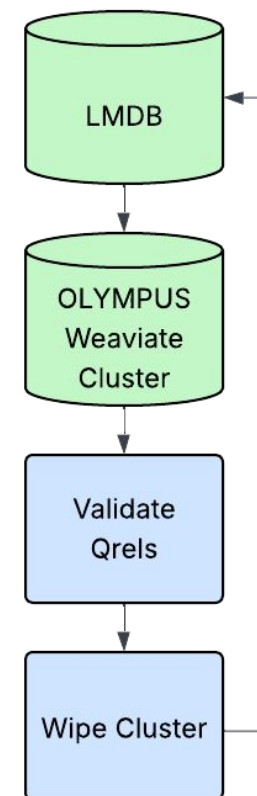
CJ Vines

Embedding Process



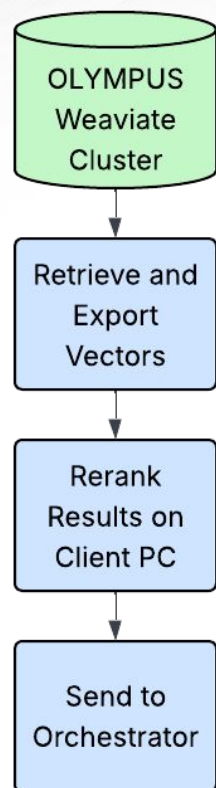
- 3 database redesigns to optimize for time and memory/space constraints (one in past 3 weeks)
- Consistent validation framework to optimize MUVERA hyperparameters
- If properly optimized, rerank not necessary. Optimization requires full corpus, so Rerank assists with nDCG if tuning is not perfect

Validation/ Upload Process



Embedding Pipeline and Storage - Retrieve/Rerank

CJ Vines



- Retrieve happens on OLYMPUS cluster which allows for larger corpus due to memory constraints
- Reranking is cheap for smaller amounts of vectors, so reranking happens on client PC
- Results given to orchestrator in less than 2 s currently at limit of 200, reranking 10 patents.

Embedding Pipeline and Storage - Current Scale/Challenges and Solutions

CJ Vines

Problems

- Memory constraints on client PC limits scale
- MUVERA representation performance can vary by design
- Hyperparameter tuning without compute takes 5 days

Solutions

- Host database on OLYMPUS cluster
- Effective tuning of hyperparameters
- Perform embedding and tuning asynchronously

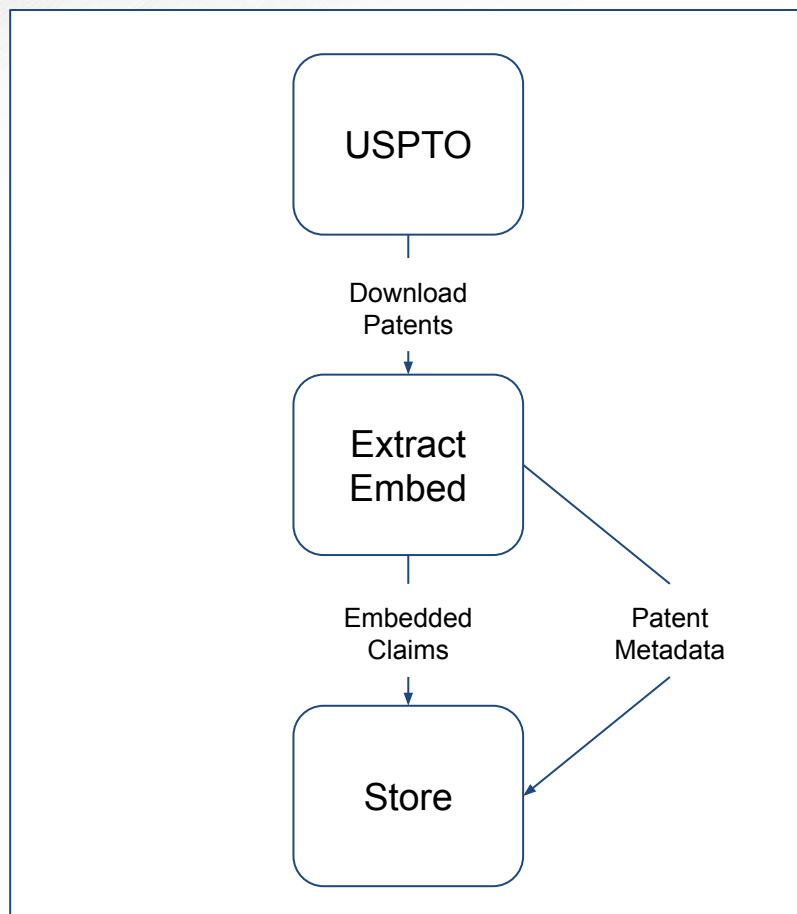
Current Scale

- Current Claim Count: 1,953,536
- Current Patent Count: 121,662
- Current Size on Scratch: 80G
- Embedding and uploading as I speak

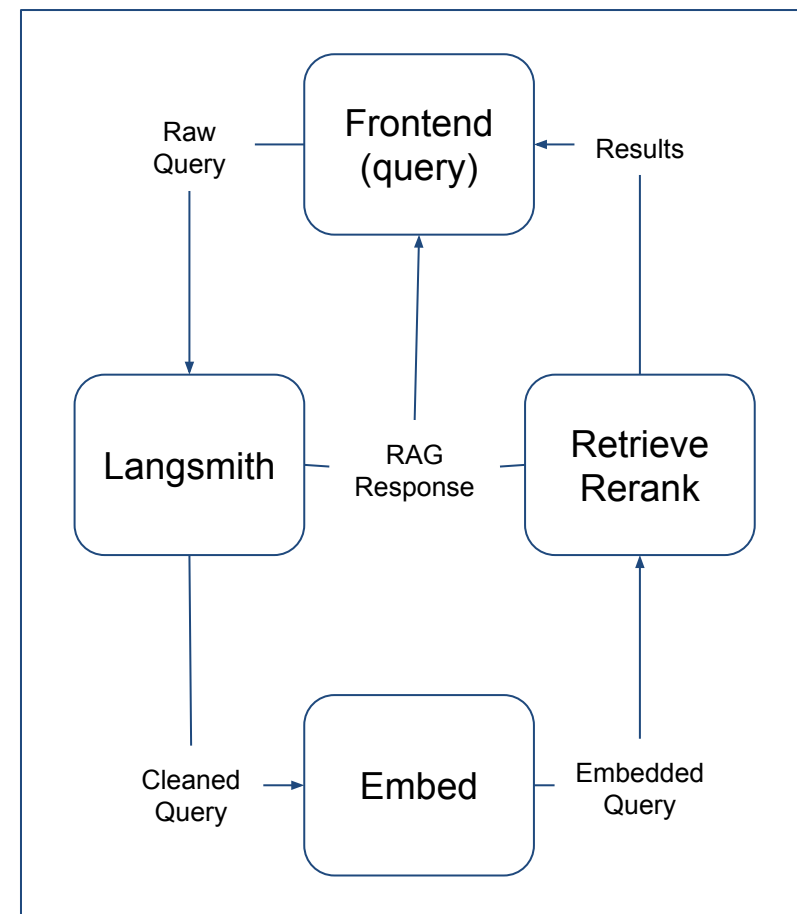
Frontend and Validation - Overview

Christian Casteel

Storage



Search



Frontend and Validation - Challenges and Solutions

Christian Casteel

Challenges:

- No precedent for validation.
- Minimum necessary patent information.

Solutions:

- Use industry standard metrics.
- Decided claims and metadata were all that were needed.

Frontend and Validation - Results

Christian Casteel

All metrics evaluated with 40 queries on 80k patent dataset

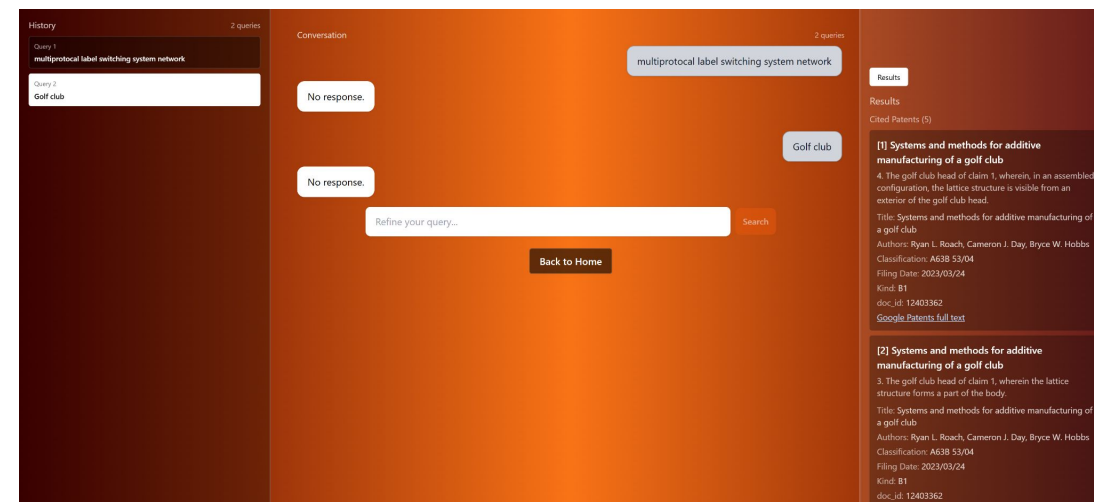
Top Results (k)	Recall@k	nDCG@k	MRR@k
10	.8	0.589	0.520
20	.825	0.596	0.522
50	.825	0.596	0.522
100	.825	0.596	0.522
200	.85	0.599	0.522

Target Accuracy Metrics

Recall@k	nDCG@k	MRR@k
.8	.4	.4

Average Search
Time (s)

6.59



Metadata Accuracy

Some metadata issues on old datasets

Full System Robustness

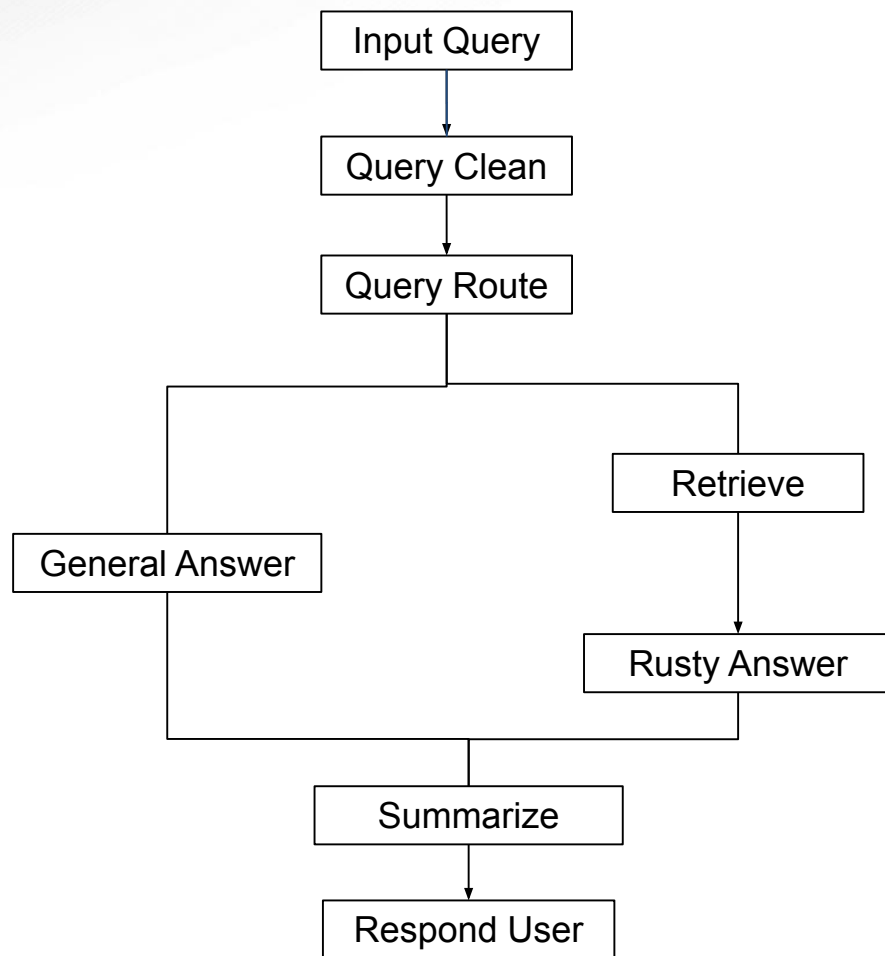
Multiple dataset ingests and queries possible in succession

Failure Compliance

System functions without LangSmith API and fails gracefully

Retrieval Augmented Generation Overview

Kosi Ezewudo



Pre-requisites:

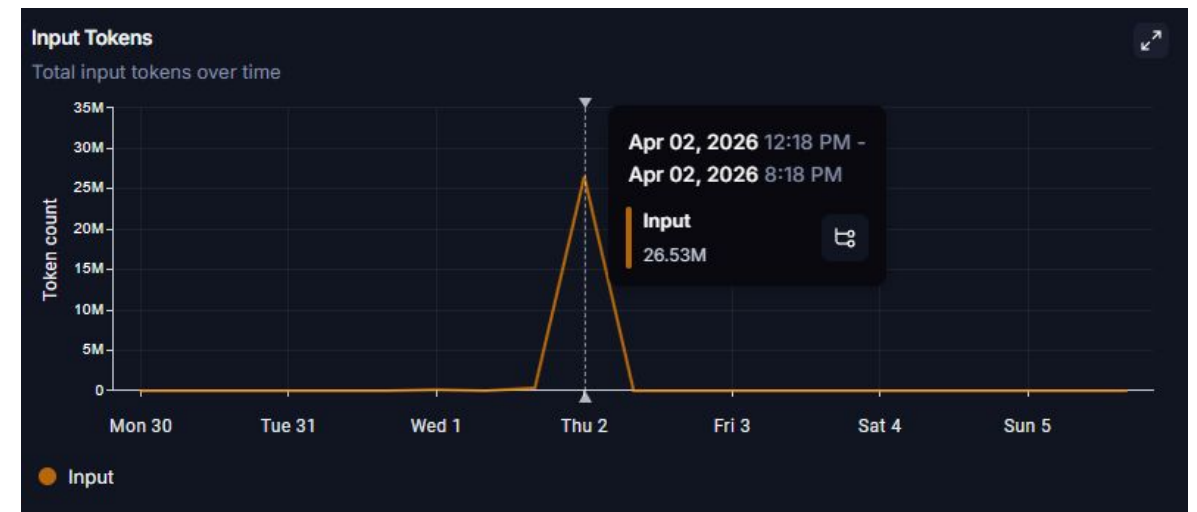
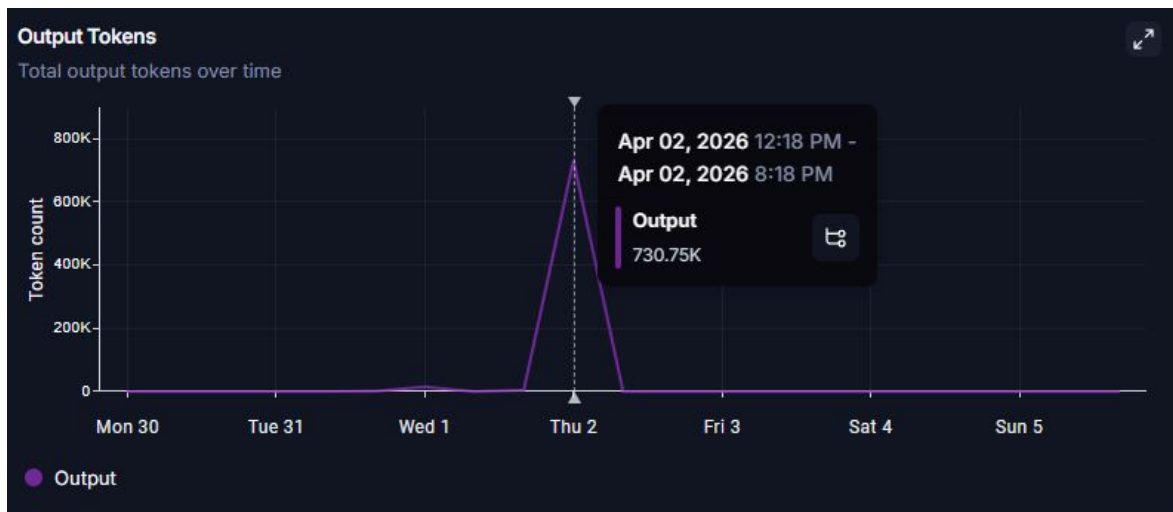
- Query cleaning & routing
- Execute context retrieval
- Conversation summary

Fig 1. Patent Miner RAG pipeline

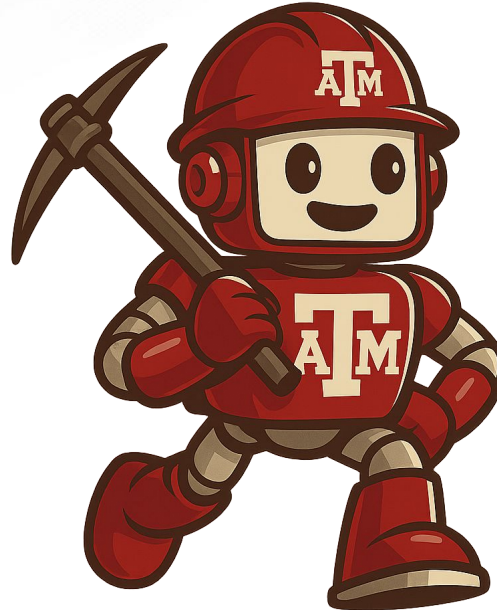
Retrieval Augmented Generation Overview

Kosi Ezewudo

Context-basis	Faithfulness	Answer relevancy	Clean Accuracy
Patent-level	.8658	0.7953	Works as expected
Chunk-level	0.1397	0.0992	Works as expected



Integrated System Results



https://youtu.be/A_Lmkt5tkHU

Current Status

Embedding/Storage:

- Tune hyperparameters
- Continue ingesting patents

Frontend/Validation:

- Fix metadata
- Filters incomplete
- Broken links for patents older than 2010s
- Validate ability to search for prior art*

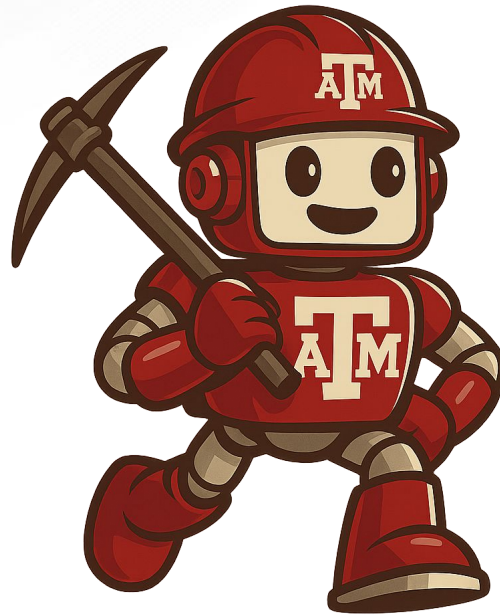
RAG/Orchestrator:

- Claim Reconstruction



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Thank you!